Kaplan-Meier survival estimates of death from any cause in patients with early-onset colorectal cancer (age < 50 yrs.) vs average-onset CRC (age ≥ 50 yrs.). Patients were followed for a maximum of 1000 days from the date of diagnosis.

Mo1209

POSITIVITY RATE AND YIELD FOR NEOPLASIA OF FECAL IMMUNOCHEMICAL TESTING (FIT) PER-ROUND OVER TIME IN ORGANIZED COLORECTAL CANCER (CRC) SCREENING PROGRAMS: SYSTEMATIC REVIEW AND META-ANALYSIS

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Background: Fecal immunochemical tests (FIT) are the most widely used tests in organized colorectal cancer (CRC) screening programs worldwide. Compared to colonoscopy or other non-invasive screening methods, FIT may be better accepted with higher participation rates, which may counteract its lower one-time sensitivities. Data on how the performance of FIT varies over successive rounds of screening have not been synthesized systematically. **Aims**: To ascertain per-round positivity rates and diagnostic yields over time for CRC, advanced adenoma (AA) and adenomas in FIT-based organized CRC screening programs. **Methods**: We performed a systematic review and meta-analysis using Pubmed, Embase and Cochrane databases. Studies of opportunistic screening were excluded. Data regarding positivity rates (positive FITs/participants) and detection rates (detected lesions/follow-up colonoscopies after positive FIT) across subsequent rounds were extracted and summary estimates and their standard errors (SE) were obtained. Yields were calculated for 2 populations: 1) Overall Population participating in the programs; 2) Consistent Screenees who completed all offered rounds. **Results, studies**: The literature search yielded 3,962 studies of which 19 were included. Table 1 shows program setting, round number, years of patient inclusion, age range, FIT cut-off and periodicity. Sparse data were available for rounds 5+. **Results, FIT-positivity:** For the Overall Population, FIT-positivity across studies from the 1st though 4th rounds was 6.7%, 5.5%, 5.3% and 5.2%, respectively. For Consistent Screenees, FIT-positivity across studies from the 1st though 4th rounds was 6.4%, 4.6%, 3.5% and 3.6%, respectively. After a positive FIT, a mean of 90% (SE 0.1) of patients underwent a follow-up colonoscopy. **Results, yield:** For the Overall Population, detection rates from the 1st though 4th rounds were: CRC 5.7%, 3.6%, 3.1% and 2.5%, respectively; AA 33.4%, 29.2%, 23.9% and 23%, respectively; adenoma 47.9%, 45.8%, 46.1% and 44.6%, respectively (Figure 1). For Consistent Screenees, detection rates from the 1st though 4th rounds were: CRC 6.6%, 4.2%, 3.4%, and 3.1%, respectively; AA 36.6%, 26.3%, 24.3% and 24.9%, respectively; adenoma 53.4%, 46.7%, 42.6% and 42.4%, respectively (Figure 2). **Sensitivity analyses:** Similar trends were observed for biennial screening with FIT cut-off of 20µg Hb/g feces. **Conclusion**: FIT-positivity, and yields for CRC, AA and adenoma are highest in the 1st round, but may stabilize at a clinically relevant steady state at rounds 3+. The steady-state FIT-positivity rate appears higher in the Overall Population vs. Consistent Screenees, but the yields for neoplasia appear similar. Additional studies are needed to clarify whether the interval between rounds could be lengthened for Consistent Screenees, which could improve the efficiency of screening programs.

First author	Publication year	Country	Years of patient inclusion	Periodicity	FIT cut-off	Number of rounds	Population age
Crotta	2012	Italy	2001-2008	Biennial	20 µg Hb/g	4	50-74
Parente	2013	Italy	2005-2009	Biennial	20 µg Hb/g	2	50-69
Bujanda	2015	Spain	2009-2013	Biennial	20 µg Hb/g	2	50-69
Portillo	2018	Spain	2009-2014	Biennial	20 µg Hb/g	3	50-69
Zorzi	2018	Italy	2002-2014	Biennial	20 µg Hb/g	6	50-69
Baldacchini	2020	Italy	2005-2016	Biennial	20 µg Hb/g	5	50-69
Tepes	2020	Slovenia	NA	Biennial	20 µg Hb/g	3	50-69
Sali	2022	Italy	NA	Biennial	20 µg Hb/g	3	54-65
Denters	2012	Netherlands	2006-2009	Biennial	10 µg Hb/g	2	50-74
Stegeman	2015	Netherlands	2007-2012	Biennial	10 µg Hb/g	3	50-75
Schreuders	2019	Netherlands	2006-2014	Biennial	10 µg Hb/g	3	50-74
Knudsen	2022	Norway	2012-2018	Biennial	>15ug Hb/g	4	50-74
Breekveldt	2022	Netherlands	2014-2017	Biennial	47 µg Hb/g	2	55-75
Van Roon	2013	Netherlands	2006-2012	1-3 years	10 µg Hb/g	2	50-74
Kapidzic	2014	Netherlands	2006-2016	1-3 years	10 µg Hb/g	3	50-74
Jensen	2016	USA	2007-2010	Annual	20 µg Hb/g	4	50-70
Choi	2012	South Korea	2004-2008	Annual	NA	5	>50
Osborne	2017	Australia	2008-2011	Annual	NA	4	50-74
Dominic	2022	USA	2012-2017	Annual	NA	6	NA

Table 1. Characteristics of included studies

FIT: fecal immunochemical test; Hb: hemoglobin

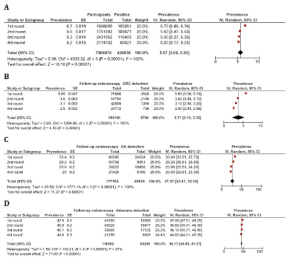


FIGURE 1. Positivity rate (A) and yield for CRC (B), AA (C) and adenoma (D) of FIT per-round in the overall screening population

CRC: colorectal cancer; AA: advanced adenoma; FIT: fecal immunochemical test; SE: standard error.

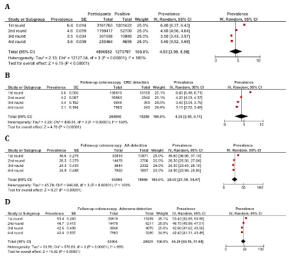


FIGURE 2. Positivity rate (A) and yield for CRC (B), AA (C) and adenoma (D) of FIT per-round in consistent screenees

CRC: colorectal cancer; AA: advanced adenoma; FIT: fecal immunochemical test; SE: standard error.

Mo1210

UNCOVERING THE RED FLAGS: A CROSS-SECTIONAL RETROSPECTIVE CASE-CONTROL STUDY ON PREDICTORS OF EARLY ONSET COLORECTAL CANCER IN A MULTI-STATE COMMUNITY HEALTH SYSTEM

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Background: The increasing prevalence and aggressive, potentially fatal, nature of early onset colorectal cancer (EOCRC) make it a critical public health concern. Identifying key risk factors and red flag symptoms is essential for facilitating earlier detection and improving patient outcomes. **Methods:** We used electronic health record data from a large community health system to identify patients diagnosed with EOCRC between 1/01/2017 and 12/31/2023 based on International Classification of Diseases (ICD)-10 codes. EOCRC cases were exact matched on age and sex with negative colonoscopy controls on a 1:1 ratio. Patients' demographics, comorbidities, hereditary syndromes, family history, symptoms, specialty visits, and medications were extracted at the time of the diagnostic visit. Data were analyzed with univariate and multivariate logistic regressions to predict the odds of EOCRC diagnosis. **Results:** We identified 3,827 EOCRC patients categorized into stages I (11%), II (19%), III (40%), and IV (29%). The mean age was 42.2 years (range 20-49), with 49% female, 70% white, and 81% non-Hispanic/Latino. Few cases were not matched and removed from analyses (N = 23). As expected, cases and controls (Ns = 3804) were equivalent on age and sex. Significant predictors from univariate logistic regressions were included in a multivariate logistic regression. This model indicated several predictors were significantly associated with higher odds of EOCRC, including Latino/Hispanic background, Medicare, Medicaid, Medicare/Medicaid dual membership, comorbidities, rectal bleeding, iron deficiency anemia, former alcohol use, metformin, and nonsteroidal anti-inflammatory drugs (NSAIDs). Several predictors were significantly associated with lower odds of EOCRC, including Asian background, marriage/domestic partnership, hyperlipidemia, total clinical visits, and vitamin d supplementation. A follow-up multivariate regression showed that all predictors remained significant, except metformin use, after including significant high-risk predictors such as hereditary syndromes and irritable bowel syndrome (IBS). **Conclusions:** Cross-sectional findings indicated Latino/Hispanic background, public insurance, comorbidities, rectal bleeding, iron deficiency anemia, former alcohol use, NSAIDs, and hereditary syndromes were associated with higher odds of EOCRC. Asian background, marriage/domestic partnership, hyperlipidemia, IBS, and vitamin d supplementation were associated with lower odds. Study strengths include the large size of EOCRC cases and control group, diversity of patients' race, ethnicity, and sex, and results based on a current cohort (2017-2023). Integrating these independent risk factors into clinical screening assessments and recommendations may facilitate earlier detection of EOCRC.

Mo1211

LARGE-SCALE PRETRAINED ARTIFICIAL INTELLIGENCE MODEL FOR POLYP DETECTION: FROM PUBLIC DATASETS TO A LOCAL RECRUITED-DATASET IN SAUDI ARABIA

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Introduction: Colonoscopy remains the gold standard for colorectal cancer (CRC) screening, particularly in regions where CRC is the most common cancer. For instance, in Saudi Arabia, CRC accounts for over 15 % of all cancers in males and 10 % in females. Despite technological advances, the accuracy of polyp detection during colonoscopy depends heavily on endoscopists' experience, highlighting the critical need for improved early detection methods. Traditional artificial intelligence (AI) approaches in colonoscopy face challenges in cross-center generalization due to limited data and population diversity, leading to decreased performance when deployed in new clinical settings or among different populations. The study aimed to develop an AI model to aid colonoscopy in polyp and cancer detection, focusing on applicability to the Saudi population. **Methods**: We developed a novel deep learning framework with self-supervised pretraining on eight public endoscopy datasets (32,941 videos, 7,266,411 frames), including 45 videos from three clinical studies within "REAL-Colon" dataset (Fig 1). The pretrained model, designed to learn robust cross-center feature representations without annotation requirements, was subsequently fine-tuned for polyp detection on "KUMC" dataset and on the same 45 videos from "REAL-Colon" dataset. Model evaluation was performed on "KUMC" test set and 15 patients from the fourth clinical study in "REAL-Colon" dataset. Additionally, we evaluated zero-shot generalization in a local cohort of 139 patients (58 with polyps, 81 normal) recruited from the King Faisal Specialist Hospital and Research Center, with a detailed quantitative analysis of 10 annotated cases. **Results:** Our model demonstrated superior performance with mean average precision (mAP) of 0.68 and F1-score of 81.2% on the "KUMC" dataset, and mAP of 0.43 and mAP50 of 0.72 on "REAL-Colon" dataset, outperforming baseline methods (mAP: 0.216, mAP50: 0.338). Remarkably, in zero-shot prediction of the local cohort, the model achieved 95.2% sensitivity, 84.3% specificity, and 88.9% accuracy, demonstrating robust generalization. Figure 2 shows the qualitative results of polyp detection. The quantitative analysis of the annotated local cases yielded 0.52 mAP, validating the model's strong transfer capability to new populations without additional training. **Conclusion:** Our large-scale pretrained AI model demonstrated exceptional generalization capabilities across different populations, particularly its zero-shot performance within the local cohort. The model's robust cross-population applicability highlights its potential for future clinical implementation in diverse geographic regions, especially in areas with high CRC prevalence. However, further validation studies involving larger and more diverse populations are essential to confirm its effectiveness in enhancing CRC screening.